r/aiwars

Environmental impact of AI

Posted by u/Bright-Accountant259 • • 0 points • 19 comments

I can't find any reliable sources of data on how harmful to the environment AI actually is and I'd rather not just blindly accept what other people have said on either side, so does anyone here have any reliable data for that?

Comments

u/Boof_That_Capacitor • 1 points • 2024-12-17 14:26 UTC

Nothing even close to when I render a 12 second clip with IRAY. Damn GPU sounds like it's going to fly away.

u/EthanJHurst • 2 points • 2024-12-17 11:29 UTC

A lot of studies have been done on this, actually. Contrary to popular belief AI actually uses *a lot* less energy for the same task than a human does.

We're **the ones that have a big environmental impact.**

u/Euchale • 1 points • 2024-12-17 11:22 UTC

Personally I don't really care for numbers from companies or papers, but I look at how the money is spent.

They are not building a nuclear power plant near their data center for "funsies" but because they expect that they will soon use a lot more electricity.

u/EffectiveNo5737 • 1 points • 2024-12-17 09:36 UTC

You can probably use crypto currency stats as a good "probably"

u/sweetbunnyblood • 2 points • 2024-12-16 22:27 UTC

Energy Use Comparison
Baldur's Gate 3 (4 Hours):
Energy Consumption: A high-end gaming PC uses about 400 watts/hour on average.

Over 4 hours: 400 watts × 4 hours = 1.6 kWh.
Carbon Footprint: Global average emissions are about 0.5 kg of CO_2 per kWh (depending on your energy source).

1.6 kWh × 0.5 kg = 0.8 kg of CO_2.
ChatGPT Usage:
A single query or response generates approximately 0.01–0.1 grams of CO_2.

For comparison, let’s assume a high-end estimate of 0.1 grams per response.
To match Baldur's Gate's 0.8 kg of CO_2 emissions:

0.8 kg = 800 grams.
800 grams ÷ 0.1 grams per query = 8,000 ChatGPT responses.
Final Comparison
Playing Baldur's Gate 3 for 4 hours on a gaming PC produces roughly the same environmental impact as 8,000 ChatGPT queries.

Perspective
If you use ChatGPT for short, text-based responses, you’d need a LOT of usage to match a single gaming session’s energy use. Gaming is significantly more energy-intensive because of the power-hungry GPUs, CPUs, and display hardware involved. ChatGPT, by comparison, is efficient for individual responses.

u/sweetbunnyblood • 1 points • 2024-12-16 22:23 UTC

Utilizing panel data from 67 countries, we employ System Generalized Method of Moments (SYS-GMM) and Dynamic Panel Threshold Models (DPTM) to analyze the complex interactions between AI development and key environmental metrics. The estimated coefficients of the benchmark model show that AI significantly reduces ecological footprints and carbon emissions while promoting energy transitions, with the most substantial impact observed in energy transitions, followed by ecological footprint reduction and carbon emissions reduction

https://www.nature.com/articles/s41599-024-03520-5

> **u/Bright-Accountant259** • 1 points • 2024-12-16 23:15 UTC
>
> This seems to be referring to AI being used as an environmental tool rather than the environmental effects of casual use of AI

u/EngineerBig1851 • 3 points • 2024-12-16 22:23 UTC

I can guarantee you that it was less than rendering Avatar.

Look into render farms and their consumption of electricity. Training a base AI model is "similiar", as in - all same hardware is used, usually in same amounts - but in a shorter timeframe (so less computational hours, and less energy consumption).

Also I have 0 idea where the water hoax even came from. Some videocards use closed cycle water cooling - but nothing in a scale of a data center or a render farm. Apparently some datacenters use "swamp cooling" (aka humidifiers), but that seems very counter intuitive when you have halls full of electrical equipment.

> **u/PM_me_sensuous_lips** • 1 points • 2024-12-16 22:48 UTC
>
> > but that seems very counter intuitive when you have halls full of electrical equipment.
>
> When you say swamp cooler, or humidifier, just think one of those towers typically associated with nuclear. That's what evaporative cooling looks like on an industrial scale (and what those towers actually do). It's not like they're going to let all the moisture out in the server room lol.

u/sweetbunnyblood • 3 points • 2024-12-16 22:21 UTC https://www.nature.com/articles/s41598-024-54271-x

14 February 2024
The carbon emissions of writing and illustrating are lower for AI than for humans

u/Cauldrath • 2 points • 2024-12-16 22:14 UTC

Keep in mind the law of conservation of energy. That electricity has to convert into something, which is almost entirely heat. If I'm running locally on my computer, that energy is also heating my home in the winter. In the summer I have a rule of no AC when I run training.

u/Phemto_B • 5 points • 2024-12-16 20:30 UTC

I can give you a source for stable diffusion at home. Each time I run it it takes about a minute. I've measured the consumption and it's about 300 watts.

Now we should compare that to if I drew something on my computer, which consumes about 70 watts just to run. I take considerably more than 5 minutes to draw something, so the AI is actually much more energy efficient.

u/Elven77AI • 12 points • 2024-12-16 20:13 UTC

Of that fearmongering was
even halfway true, gamers would waste an order of magnitude more per day:
think of it, the generation of content via GPU is both
less intensive than a typical 3D engine running and is not constant load(unless you batch
process images all day).
The cost in training the model depend on its architecture, but
its a one-time cost - not a constant strain 24/7 unlike e.g. Bitcoin miners.

u/sporkyuncle • 2 points • 2024-12-16 19:07 UTC

Regarding personal use at home...I don't have a "source" for this per se, but it's simple math that you can read up on anywhere:

For simplicity of calculation, a modern powerful computer might have a 1000 watt power supply in it. If it was used at full power for one hour, that would be 1 kilowatt hour (kWh). In the US, typical rates range from $0.10 to $0.30 per kWh.

If it takes 5 seconds to generate a 1024x1024 image in SDXL with an NVidia 4090 graphics card, this works out to 720 images per hour, which (with aforementioned full power utilization) might cost about 15 cents to make. Similarly, playing a video game on this computer for one hour with full utilization would also cost about 15 cents.

But a computer practically never runs at full power like this, even while generating or gaming. I wouldn't know what to estimate that it usually runs at, maybe 50-60%. You can also perform AI generation on lesser systems, for example an NVidia 3060 graphics card with associated components might be just fine with a 650 watt power supply, which would max out at only 65% of the kWh price ($0.07 to $0.20 per hour). Of course this would also generate fewer images.

A 1000 watt computer running full blast for an entire month of 31 days would cost $74.40 to $223.20, depending on where you live. In practice? Probably nowhere near that much. Maybe it tends to operate at about 60% utilization for the 6 hours a day you're using it, which would be more like $11.16 to $33.48 per month.

So, whatever associated environmental impact of that amount of electricity usage would be.

u/Bill3463 • 6 points • 2024-12-16 18:25 UTC

Another source for impact: https://arxiv.org/abs/2309.14393

The impact of training is probably not that large compared to other human activities. I would be worried more about the impact associated with running those models.

But also there are also new methods on the horizon that can possibly reduce the impact significantly: https://arxiv.org/abs/2402.17764

Certainly there will be more research into this domain since lower resource usage will mean lower cost for operations.

u/Hugglebuns • 2 points • 2024-12-16 18:09 UTC

It would probably pay to individually look at the environmental cost of training, and the environmental cost of rendering. That and local models vs cloud-server farm models have different rendering impacts. I don't think there is a good single number you can point to as much as the wide variety of forms that exist out there and trying to piece things together.

It doesn't help that AI is a rather large bucket, artgenAIs are a relative niche if we assume most ML projects have been labelled AI due to marketing. It might be hard to separate the cancer detectors from the chatbots so to speak

u/Mataric • 18 points • 2024-12-16 17:59 UTC

While I can't provide accurate data on the training, I can tell you that many people completely misunderstand the data and assume that every image generated uses the insanely high costs the clickbait articles have stated. This is absolutely not the case.

On my home PC (which is nothing special), it takes me a few seconds to generate an image. My electric bill is not impacted any more than it would be for playing a newer video game for those same few seconds.

u/ninjasaid13 • 0 points • 2024-12-16 17:50 UTC

https://preview.redd.it/u844zxvn097e1.png?width=907&format=png&auto=webp&s=36d835eabd0654bcaca810007622efd67b7fe4f8

I've seen this graph before, but I'm not sure how accurate this is.

> **u/PM_me_sensuous_lips** • 2 points • 2024-12-16 18:56 UTC
>
> it's from the pixart-alpha paper.